

Lab7 Setup



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Add the second form

1. Open the Project menu.
2. Click Add Form (Windows Forms)...
3. Keep the name Form2.vb. Click Add.
4. Solution Explorer now shows Form1.vb and Form2.vb.
5. Double-click a file name to open that form.
6. Put the controls on Form2.

A screenshot of a Windows Forms application window titled 'Form2'. The window has a standard Windows title bar with minimize, maximize, and close buttons. The form contains a 'Quantity' label and a text box. Below it is a 'Results' label and a group box containing three text boxes labeled 'Min', 'Max', and 'Mean'. To the right of the form are three buttons: 'Analyze', 'Exit', and 'Go to Write'.

Form2 needs:

4 Labels, 1 GroupBox, 4 TextBoxes, 3 Buttons

Lab7 Setup



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Name the text boxes

Click a text box. Find (Name) in the Properties window. Type the name below.

Form1

Quantity	txtQuantity
Low	txtLow
High	txtHigh
Status	txtStatus

Status: also set
Multiline = True

Form2

Quantity	txtQuantity
Min	txtMin
Max	txtMax
Mean	txtMean

Add the code

Double-click a button. Visual Studio creates an empty Sub. Type the code from the slide inside that Sub.

Lab7 Form1



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Form1

Quantity

Status

Range

Low

High

Generate Data Save

Go to Read Exit

Text box names on Form1

Quantity	txtQuantity
Low	txtLow
High	txtHigh
Status	txtStatus

Type under Public Class Form1:

```
Dim Num() As Integer  
Dim n As Integer
```

Generate Data

Double-click the button. Type inside the Sub:

```
Dim Low, High, i As Integer  
Dim message As String
```

```
' Get the values from the text boxes  
n = CInt(txtQuantity.Text)  
Low = CInt(txtLow.Text)  
High = CInt(txtHigh.Text)
```

Note: Num() and n stay outside the Sub. The Save button uses them too.

Lab7 Form1



TEXAS A&M
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Form1

Quantity

Status

Range

Low

High

Generate Data Save

Go to Read Exit

Text box names on Form1

Quantity	txtQuantity
Low	txtLow
High	txtHigh
Status	txtStatus

Generate Data

Continue in the same Sub:

```
' Set the size of the array
ReDim Num(0 To n - 1)

' Make the random numbers
Randomize()
For i = 0 To n - 1
    Num(i) = CInt(Rnd() * (High - Low) + Low)
Next
```

Note: The first index is 0. The last index is n - 1.

Lab7 Form1



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Form1

Quantity

Status

Range

Low

High

Generate Data Save

Go to Read Exit

Text box names on Form1

Quantity	txtQuantity
Low	txtLow
High	txtHigh
Status	txtStatus

Generate Data

Continue in the same Sub:

```
' Show 10 numbers in each line
message = ""
For i = 0 To n - 1
    If (i Mod 10) = 0 Then
        message = message & vbCrLf & CStr(Num(i))
    Else
        message = message & vbTab & CStr(Num(i))
    End If
Next

' Display the message in the text box
txtStatus.Text = message
```

Lab7 Form1



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Form1

Quantity

Status

Range

Low

High

Generate Data Save

Go to Read Exit

Text box names on Form1

Quantity	txtQuantity
Low	txtLow
High	txtHigh
Status	txtStatus

Save

Double-click the button. Type inside the Sub:

```
Dim i As Integer
```

```
' Open the file for writing
FileOpen(1, "data.txt", OpenMode.Output)
' Write 10 numbers in each line
For i = 0 To n - 1
    If ((i + 1) Mod 10) = 0 Then
        Write(1, Num(i))
        WriteLine(1)
    Else
        Write(1, Num(i))
    End If
Next
' Close the file
FileClose(1)
' Notify the user
MsgBox("Data saved.")
```

Note: The file is in the project folder: bin\Debug

Lab7 Form1



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Form1

Quantity

Status

Range

Low

High

Generate Data Save

Go to Read Exit

Text box names on Form1

Quantity	txtQuantity
Low	txtLow
High	txtHigh
Status	txtStatus

Go to Read

Double-click the button. Type inside the Sub:

```
Me.Hide()  
Form2.Show()
```

Exit

Double-click the button. Type inside the Sub:

```
End
```

Note: Always close the program with the Exit button.

Lab7 Form2



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Text box names on Form2

Quantity	txtQuantity
Min	txtMin
Max	txtMax
Mean	txtMean

Analyze

Double-click the button. Type inside the Sub:

```
Dim n, num, sum, min, max As Integer
Dim mean As Decimal
Dim data As String = ""

' Open the file for reading
FileOpen(1, "data.txt", OpenMode.Input)
```

Note: Use the same file name as in Form1.

Lab7 Form2



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Form2

Quantity

Results

Min

Max

Mean

Analyze

Exit

Go to Write

Text box names on Form2

Quantity	txtQuantity
Min	txtMin
Max	txtMax
Mean	txtMean

Analyze

Continue in the same Sub:

```
' Read the first number
```

```
Input(1, data)
```

```
num = CInt(data)
```

```
' Set the starting values
```

```
n = 1
```

```
sum = num
```

```
max = num
```

```
min = num
```

Lab7 Form2



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Form2

Quantity

Analyze

Results

Min

Max

Mean

Exit

Go to Write

Text box names on Form2

Quantity	txtQuantity
Min	txtMin
Max	txtMax
Mean	txtMean

Analyze

Continue in the same Sub:

```
' Read the rest of the data
Do Until EOF(1)
    Input(1, data)
    ' Skip an empty item
    If data <> "" Then
        num = CInt(data)
        If num > max Then
            max = num
        End If
        If num < min Then
            min = num
        End If
        sum = sum + num
        n = n + 1
    End If
Loop
```

Note: Each line in the file ends with a comma. It gives an empty item.

Lab7 Form2



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Form2

Quantity

Results

Min

Max

Mean

Analyze

Exit

Go to Write

Text box names on Form2

Quantity	txtQuantity
Min	txtMin
Max	txtMax
Mean	txtMean

Analyze

Continue in the same Sub:

```
' Close the file
```

```
FileClose(1)
```

```
' Calculate the mean
```

```
mean = sum / n
```

```
' Display the results
```

```
txtQuantity.Text = CStr(n)
```

```
txtMin.Text = CStr(min)
```

```
txtMax.Text = CStr(max)
```

```
txtMean.Text = CStr(mean)
```

Lab7 Form2



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Text box names on Form2

Quantity	txtQuantity
Min	txtMin
Max	txtMax
Mean	txtMean

Go to Write

Double-click the button. Type inside the Sub:

```
Me.Hide()  
Form1.Show()
```

Exit

Double-click the button. Type inside the Sub:

```
End
```

Note: Always close the program with the Exit button.

Lab7 Test



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Test your program

1. Press F5.
2. Type Quantity 25, Low 1, High 100.
3. Click Generate Data, then Save.
4. Click Go to Read, then Analyze.
5. Quantity shows 25. Min is 1 or more. Max is 100 or less.
6. Click Exit.

Remember

- Fill all three text boxes with whole numbers.
 - Click Generate Data before Save.
 - Click Save before Analyze.
 - The text box names must match the code.
 - Close the program with the Exit button.
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